METRIC

A-A-54733
11 September 1992

COMMERCIAL ITEM DESCRIPTION

GRADUATE, LIQUID, LABORATORY, PLASTIC

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-G-36805 which is cancelled.

This Commercial Item Description covers one type of plastic double scale graduates for pharmaceutical use.

Salient characteristics:

General. Shall be plastic pharmaceutical graduates having both metric (ml) and inch-pound (fluid oz) scales. The graduates shall be of the following double scale capacity sizes (see "ordering data"):

Size 1. - 125 ml and 4 oz.
Size 2. - 250 ml and 8 oz.
Size 3. - 500 ml and 16 oz.
Size 4 - 1000 ml and 32 oz.

The graduates shall be conical and rigid, and shall include a pour-out spout and an octagonal base. The pour-out spout shall permit the delivery of an unbroken stream of liquid without it wetting the outside of a graduate. The base shall be perpendicular to vertical axis of its graduate and shall be large enough for the empty graduate to stand and not fall when placed on a surface inclined 15° from the horizontal. The interior of the graduates shall have smooth surfaces.

AMSC N/A FSC 6640

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Material. The graduates shall be fabricated from transparent shatterproof plastic that is either polypropylene or polymethylpentene. The transparent plastic shall not obstruct the viewing of a liquid level from the outside of a graduate. The plastic graduates shall have sufficient heat resistance to withstand a 121°C autoclaving temperature for 20 minutes without deforming, cracking, or becoming opaque.

Graduation markings. The graduates shall have two adjacent scales with a vertical scale separation line. One scale shall be in milliliters and the second scale shall be in fluid ounces. The graduation markings shall be raised and legible. The graduation markings shall be calibrated to deliver (TD) the indicated volume at 20°C. The inside measurement, within a graduate, from the bottom to a graduation marking shall not be less than twice the inside diameter at that graduation marking. The double scale marking requirements for the graduation ranges, graduation subdivisions and numbering shall be as indicated in table I. The designation for the units for the double scale may be abbreviated. The main graduation lines shall be larger than the subdivisions, and shall be numbered with Arabic numerals.

Table 1. Double Scale Graduation Markings

Designated Size	Capacity (Ounces)	Graduated Range (Ounces)	Subdivisions (Ounces)	Main and Numbered Graduation Lines (Ounces)
1	4	1 to 4	1/2	1
2	8	2 to 8	1/2	1
3	16	4 to 16	1	2
4	32	8 to 22	1	4
	(ml)	(ml)	(ml)	(ml)
1	125	25 to 125	5	25
2	250	50 to 250	10	50
3	500	100 to 500	25	100
4	1000	200 to 1000	25	1000

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Volumetric accuracy. The graduates shall be calibrated to deliver (TD) their indicated volumes at 20°C. The volumetric accuracy of a specific graduation line in the metric scale shall be as indicated in table II. The volumetric accuracy of a graduation line shall be based on the following gravimetric testing procedure:

Testing conducted under conditions where the room temperature is 20°C ±5°C.

A graduate that is thoroughly clean is filled with distilled water, at room temperature, to the level of the graduation line.

The water in the graduate shall be poured into a tared container whose weight has been recorded.

The weight of the filled container shall be determined and recorded, as well as the temperature of the water which shall be measured and recorded.

The weight of the empty container shall be subtracted from the weight of the filled container to determine the weight of the water sample.

The volume of the water sample shall be calculated by dividing the weight of the water sample by the relative density of the water at its testing temperature, based on using a standard reference that gives the relative density of distilled water at different temperatures.

Table II. Volumetric Accuracies at 20°C.

Inside Diameter at Graduation Line (mm)	Volumetric Accuracy ± ml
21 to 26	0.4
26 to 31	0.6
31 to 36	0.8
36 to 41	1.1
41 to 46	1.4
46 to 51	1.8
51 to 56	2.2
56 to 61	2.8
61 to 66	3.4
66 to 71	4.1
71 to 76	4.8
76 to 81	5.6
81 to 86	6.4

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Identification marking. Each graduate shall include raised marking that indicates the manufacturer's name or registered trademark, the calibration means (to deliver), and temperature of 20°C. The calibration means and temperature references may be abbreviated.

Workmanship. The graduates shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each (EA). One graduate, of the size specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination and Testing. Examination and testing shall be conducted on the graduates in accordance with commercial practice. The sampling for examination and testing shall be in accordance with commercial practice.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. (Capacity size, intermediate/exterior package quantities, labeling and marking must be specified in the contract and/or order).

NOTE: The following National Stock Numbers are covered by this document:

NSN	SPECIFICATION DESIGNATED FACF	Item Identification
6640-00-889-7083	1	GRADUATE LIQUID, LABORATORY, Plastic 125 ml.
6640-00-889-7084	2	GRADUATE LIQUID, LABORATORY, Plastic 250 ml.
6640-00-889-7085	3	GRADUATE LIQUID, LABORATORY, Plastic 500 ml.
6640-00-889-7086	4	GRADUATE LIQUID, LABORATORY, Plastic 1000 ml.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6640-1750

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER A-A-54733	2 DOCUMENT DATE (YYMMDD) 11 SEPTEMBER 1992

3 DOCUMENT TITLE
GRADUATE, LIQUID, LABORATORY, PLASTIC

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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